

Macro Seasons v4

Data sources, season rules and monthly portfolio construction.

At each month-end, the model calculates one Growth score, one Inflation score and one Liquidity score. Growth and Inflation set the four season weights. Liquidity changes the final mix between risk and defensive ETFs.

1 The four seasons

Season	Growth and inflation	Economic reading	Fixed sleeve emphasis
SPRING	Growth rising Inflation cooling	Recovery with disinflation	Broad equities, technology, credit, housing and modest duration
SUMMER	Growth rising Inflation heating	Reflationary expansion	Cyclicals, value, energy, materials, emerging markets, commodities and TIPS
FALL	Growth falling Inflation heating	Stagflation	Gold, broad commodities, inflation protection, dollar, defensive equities and short duration
WINTER	Growth falling Inflation cooling	Disinflationary slowdown	Treasuries, defensive and low-volatility equities, gold, safe-haven currencies and cash

2 Data used for each pillar

GROWTH Data and source ALFRED: Industrial Production (INDPRO) and Nonfarm Payrolls (PAYEMS). FRED: Initial Jobless Claims (ICSA). Yahoo Finance: XLI, XLP, SPY and IEF adjusted prices. Changes used: industrial production over 6m; payrolls over 3m; claims over 3m (falling claims count as stronger growth); XLI versus XLP and SPY versus IEF over 6m.	INFLATION Data and source ALFRED: Consumer Price Index (CPIAUCSL). FRED: Five-Year Breakeven Inflation (T5YIE) and WTI crude oil (DCOILWTICO). Yahoo Finance: TIP and IEF adjusted prices. Changes used: change in 12-month CPI inflation over 3m; breakeven inflation over 3m; WTI and TIP versus IEF over 6m.	LIQUIDITY Data and source ALFRED: M2 money stock (M2SL). FRED: NFCI, high-yield OAS, 2-year Treasury yield, Fed balance sheet, reverse repos and the Treasury General Account. Changes used: NFCI level and 3m change; OAS 3m; M2 6m; 2-year yield 12m; Fed net liquidity 3m. Portfolio effect: easier liquidity can move up to 15 points from defensive ETFs to risk ETFs; tighter liquidity can move up to 25 points the other way.
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3 From month-end data to next month's portfolio

1	Select the month-end data	Use ALFRED decision-date values for CPI, industrial production, payrolls and M2. Apply the coded release delay to the other FRED series, which use latest-revised history. Use Yahoo adjusted prices through month-end.
2	Calculate three scores	Calculate the listed trailing changes and set their signs so positive means stronger growth, higher inflation or easier liquidity. Standardize each reading against up to 120 prior months, with at least 36 months required and values capped at three standard deviations. Average the available readings and apply a two-month exponential average.
3	Calculate four season weights	Stronger growth raises Spring and Summer; weaker growth raises Fall and Winter. Higher inflation raises Summer and Fall; lower inflation raises Spring and Winter. The four values total 100% and are used directly as sleeve weights. They classify current conditions rather than forecasting event probabilities.
4	Build next month's holdings	Adjust each manually specified season template with trailing volatility, blend all four templates using the season weights, then apply liquidity, real-yield, credit, momentum, trend and portfolio-volatility rules in that order.
5	Record next month's return	Apply the month-end weights to the following month's returns. The long-only backtest charges 10 basis points per unit of turnover. The later physical L/S simulation applies its separate trading and financing assumptions.

4 ETF templates and selection record

Starting template weights in V3/V4 Spring: SPY 20%, QQQ 15%, SMH 10%, IWM 10%, XHB 7.5%, HYG 15%, LQD 7.5%, IEF 10%, GLD 5%. Summer: XLE 15%, XLB 10%, XLI 10%, VLUU 10%, IWM 10%, EEM 10%, DBC 12.5%, CPER 5%, GLD 7.5%, TIP 10%. Fall: GLD 17.5%, DBC 10%, XLE 7.5%, TIP 15%, UUP 10%, XLP 10%, XLV 7.5%, XLV 7.5%, SHY 7.5%, BIL 7.5%. Winter: TLT 17.5%, IEF 17.5%, SHY 7.5%, GLD 12.5%, USMV 10%, XLP 10%, XLV 10%, FXY 5%, FXF 5%, BIL 5%.	What the research record supports The June prototype used a historical Sharpe optimizer. V2 replaced those portfolios with the literal template weights shown here. V3 copied the V2 templates unchanged, every H1-H8 variant retained them, and V4 calls the V3 portfolio functions. No V2-V4 optimizer estimates these constituents or starting weights. The available files do not contain a contemporaneous decision memo for every ETF and weight, or independent proof that the templates were chosen without inspecting historical returns. They are therefore described as manually specified macro templates.
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Research record and portfolio rules

What changed from the June optimizer to V4, followed by the exact long-only and physical L/S construction steps.

5 Version history

JUNE PROTOTYPE

The standalone research script ranked a broad asset universe by season and used a constrained Sharpe optimizer to choose holdings and weights. Its portfolios were discarded and do not feed V3/V4.

V2 REDESIGN

V2 introduced the current symmetric Growth x Inflation seasons, separate Liquidity overlay and manually coded ETF templates. It also introduced the trailing inverse-volatility adjustment and soft four-season blend.

V3 RESEARCH

V3 did not change ETF membership or starting weights. Eight variants, H1 through H8, tested overlays and portfolio construction on returns through December 2018. V3 retained H8 real yields, H3 credit, H2's 200-day trend gate and H5's three-stream combination.

V4 AND EXECUTION

V4 kept the V3 portfolio functions and rebuilt the data contract with four ALFRED histories and freshness gates. The physical IBKR L/S translation was added afterward; it changes execution and cost accounting, not the macro signal.

6 Current portfolio machinery

LONG-ONLY PORTFOLIO

Weights sum to 100%. The portfolio holds long ETF positions and BIL. It has no shorts or borrowing.

1. Template adjustment. For each season, start from the coded weights in section 4. Calculate trailing 36-month volatility, form a version that divides each starting weight by its volatility, and average that version 50/50 with the starting weights.
2. Season and liquidity blend. Combine all four adjusted templates using the Growth/Inflation season weights. The Liquidity score can move up to 15 percentage points from defensive ETFs to risk ETFs, or up to 25 points from risk to defense.
3. Real yields and credit. The six-month change in DGS10 minus T5YIE rotates up to half of the available GLD or TLT/IEF sleeve. Credit stress is active when the selected spread exceeds 110% of its trailing 36-month median and has widened over three months; the model then halves the risk sleeve.
4. Momentum and trend. Rank current non-BIL holdings by return from 12 months ago to one month ago; the rank multiplier ranges from 0.75 to 1.25. Move each ETF below its 200-day moving average to BIL, using the 10-month average only when daily history is insufficient.
5. Portfolio volatility and cost. Estimate the current weight vector's volatility from the trailing 24 monthly returns. Scale can range from 0.50 to 1.00 around the 10% target; unused weight goes to BIL. Charge 10 basis points per unit of turnover.

This is the funded portfolio shown in the allocation table.

L/S PORTFOLIO

The later execution overlay converts the V3 three-stream ensemble into ETF positions, nets exposures and applies the stored IBKR cost assumptions.

1. Two season streams. The core stream uses the season blend and Liquidity overlay. The enhanced stream uses the complete long-only process. Each stream is separately scaled between 0.50 and 1.50 using volatility through the previous month.
2. Time-series momentum stream. The 13 ETFs are SPY, QQQ, IWM, EFA, EEM, TLT, IEF, GLD, DBC, UUP, FXY, FXF and HYG. Each is long when its trailing 12-month return exceeds BIL and short otherwise.
3. Momentum sizing. Divide the 10% position-volatility numerator by trailing 36-month volatility and by the 13-asset universe size. Cap each position at 20% of that stream. The stream itself is scaled between 0.50 and 1.50 toward 10% volatility.
4. Combine streams. Weight the three streams inversely to their trailing 24-month volatility, with the window shifted one month. Scale the combined return stream between 0.50 and 1.50 toward 10% volatility.
5. Execution translation. Add duplicate positions, cancel opposing exposures, reduce BIL before recording a USD debit, and apply the configured gross, net, short, asset and borrowing limits. The simulator then charges the dated settings in the execution CSV files.

This execution layer was added after the V4 signal freeze.

7 Evidence and implementation boundaries

V3 development record

The variant scripts and Config A/B summaries use return dates through December 2018. The freeze document reports about 60 configurations, followed by a final Config A/B comparison, and records a one-time January 2019-May 2026 evaluation after Config A was selected. The first Git commit containing that source and freeze record is dated August 20, 2026, so Git does not independently prove the earlier pre-registration sequence.

V4 data release

Commit d30e26e froze the PIT data release on August 20, 2026 with data through July 31 and an August allocation. Its reconstructed history is retrospective. The recorded first post-freeze signal is August 31 and the first post-freeze monthly return is September 30.

Physical L/S overlay

Commit 3b8e39b added the IBKR execution translation later on August 20. Its January 2007-July 2026 history was simulated after the V3/V4 strategy had been developed. It is execution research, not an untouched holdout or a live account record.

Long-only limits

ETF and BIL weights total 100%. Exposure can fall as low as 50% when volatility is high; the remainder goes to BIL. Exposure cannot exceed 100%. Shorts and margin borrowing are disabled.

L/S limits

Maximum gross exposure: 175%. Maximum net exposure: 150%. Maximum gross shorts: 35%. Maximum absolute weight per non-cash ETF: 25%. Maximum USD margin debit: 50%. IBKR calculates the final Portfolio Margin requirement on the live account.

Dated execution assumptions

The August 20, 2026 configuration uses \$0.005 per share, a \$1 order minimum, US regulatory-fee rates, 1 basis point of slippage and stored margin and short-proceeds tiers. It assumes 1% annual borrow for every short ETF because historical borrow data is unavailable. Taxes and market impact are excluded.

Historical data contract

ALFRED decision-date histories are used for CPIAUCSL, INDPRO, PAYEMS and M2SL. The remaining FRED inputs use coded publication delays applied to latest-revised histories. Yahoo adjusted prices provide traded returns. Mutual funds and gold futures extend selected ETF histories before inception.

Known V4 data issue

The Liquidity code subtracts RRPONTSYD directly from WALCL and WTRGEN. FRED reports RRPONTSYD in billions of dollars and the other two in millions, but V4 does not rescale RRP before subtraction. This understates the reverse-repo term. Correcting it changes the frozen signal and requires a new model version.